

PHYS 253: Thermal Physics

W.R.YCH.

August 31st, 2026

Abstract

NOT FINISHED

Contents

1	Introduction	1
2	Energy Conservation	1
3	Thermal Equilibrium	1
4	Entropy	1
A	Appendix	1
B	Useful Links	1

1 Introduction

Textbook used is An Introduction to Thermal Physics by Daniel V. Schroeder. Taught by Bradley Siwick. Mondays, Wednesdays and Fridays from 8:35 to 9:25 in rutherford 118. Course grading is divided as follows:

- 10% problem sets (6, must be handwritten pdf scans) ¹.
- 50% 2 midterms.
- 40% final exam.

August 31,
2026

Definition 1 (Thermodynamics). *Thermodynamics, derived from the ancient Greek words *therme* (heat) and *dynamis* (power), is the branch of physics that deals with the relationships between heat, work, temperature, and energy.*

Remark 1. *Thermodynamics does not give microscopic details of the system², but rather describes the macroscopic properties and behaviors of systems in terms of a few state variables.*

2 Energy Conservation

3 Thermal Equilibrium

4 Entropy

A Appendix

B Useful Links

¹Only one or two of the problems on each assignment will be indicated for submission and grading.

²Classical thermodynamics cannot describe microscopic systems directly, but modern extensions like stochastic thermodynamics and statistical mechanics can.